

# ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR E-COMMERCE PLATFORM TRANSFORMATION

## DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

### 1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

### 2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

### 3. OPERATING MODEL OVERVIEW

#### 3.1. Governance Model Type

##### ✓ Federated Governance Model

- Central governance defines policies & standards
- Domain teams execute governance within business units

### 4. GOVERNANCE STRUCTURE

#### 4.1. Organizational Layers

##### Executive Layer

- Chief Data Officer (CDO)
- Data Governance Council

##### Strategic Layer

- Data Governance Lead
- Domain Data Owners

### Operational Layer

- Data Stewards
- Data Custodians (IT)

### Supporting Functions

- Risk & Compliance
- Legal
- Security

## 4.2. Roles & Responsibilities

| Role               | Responsibility                                   |
|--------------------|--------------------------------------------------|
| Data Owner         | Accountable for data domain quality & compliance |
| Data Steward       | Ensures data quality, metadata, definitions      |
| Data Custodian     | Technical management of data                     |
| Governance Council | Decision-making & policy approval                |

## 5. RACI FRAMEWORK

| Activity                | Owner | Steward | Custodian | Governance Council |
|-------------------------|-------|---------|-----------|--------------------|
| Data Definition         | A     | R       | C         | I                  |
| Data Quality Monitoring | A     | R       | C         | I                  |
| Access Control          | A     | C       | R         | I                  |
| Policy Approval         | R     | C       | I         | A                  |
| Metadata Management     | A     | R       | C         | I                  |
| Issue Resolution        | A     | R       | C         | I                  |

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

## **6. CORE GOVERNANCE PROCESSES**

### **6.1. Data Quality Management**

- Define critical data elements (CDE)

Determination criteria:

- Regulatory/reporting impact
- Revenue/payment impact
- Customer experience impact
- Fraud/risk impact
- Operational dependency across domains

Example of e-commerce CDE: customer\_id, email, order\_id, payment\_status, sku, vendor\_id.

- Establish data quality rules & thresholds
- Monitor via dashboards
- Issue management workflow:
  - Detect → Log → Assign → Resolve → Validate → Close

### **6.2. Metadata Management**

- Maintain enterprise data catalog
- Define business glossary
- Track data lineage (source → transformation → consumption)

### **6.3. Data Access & Security**

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC) (optional advanced)
- Access request & approval workflow
- Periodic access review

### **6.4. Data Lifecycle Management**

- Define lifecycle stages:
  - Create → Store → Use → Archive → Delete
- Retention policies based on regulatory requirements

## **6.5. Data Classification**

- Define classification levels:
  - Public
  - Internal
  - Confidential
  - Restricted (PII / highly sensitive)

## **7. GOVERNANCE ARTEFACTS**

### **Mandatory Artefacts:**

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

## **8. TECHNOLOGY & TOOLING**

### **8.1. Data Governance Tools**

- Data Catalog: Collibra / Alation / Atlan / Microsoft Purview
- Data Quality: Informatica / Great Expectations

### **8.2. Data Platforms**

- Data Warehouse (Snowflake, BigQuery)
- Data Lake / Lakehouse (Databricks)

### **8.3. Integration**

- Metadata ingestion from ETL pipelines
- Automated lineage tracking

## 9. INTEGRATION WITH DATA ARCHITECTURE

- Governance embedded in:
  - Data ingestion pipelines
  - Transformation layers
  - BI / reporting layer
- Controls applied at:
  - Data creation
  - Data storage
  - Data access

## 10. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

## 11. KPIs & SUCCESS METRICS

| Metric                   | Description                                                                |
|--------------------------|----------------------------------------------------------------------------|
| Data Quality Score       | >95% of data meeting quality standards                                     |
| Issue Resolution SLA     | >95% of issues are resolved within the SLA according to severity           |
| Data Availability        | >99.5% availability for Tier-1 datasets during business hours / agreed SLA |
| Metadata Completeness    | >98% of critical datasets documented in the catalog                        |
| Access Review Completion | 100% quarterly completion for privileged roles                             |

**KPI Definitions:**

| KPI                      | Formula (Example)                                                               | Target Phase 1–2 | Target Steady State |
|--------------------------|---------------------------------------------------------------------------------|------------------|---------------------|
| DQ Score                 | % CDE records that pass all quality rules                                       | ≥90%             | ≥95%                |
| Metadata Completeness    | % critical data assets with owner, steward, definition, classification, lineage | ≥80%             | ≥98%                |
| Owner Coverage           | % critical data assets with Data Owner                                          | ≥95%             | 100%                |
| Issue Resolution SLA     | % of issues are resolved within the SLA according to severity                   | ≥80%             | ≥95%                |
| Access Review Completion | % quarterly access reviews completed on time                                    | ≥90%             | 100%                |

**12. IMPLEMENTATION ROADMAP****Phase 1: Assessment**

- Current state analysis
- Gap identification

**Phase 2: Design**

- Define TOM
- Develop policies & standards

**Phase 3: Pilot**

- Implement in selected domain

**Phase 4: Rollout**

- Scale across enterprise

**Phase 5: Optimization**

- Continuous improvement

### 13. CHANGE MANAGEMENT & ADOPTION

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

### 14. ESCALATION FRAMEWORK

- Data issues escalated:
  - Steward → Owner → Governance Council
- SLA for resolution defined

SLA Table (Example):

| Severity | Initial Response | Target Resolution |
|----------|------------------|-------------------|
| Critical | < 4 jam          | < 24 jam          |
| High     | < 1 hari kerja   | < 5 hari kerja    |
| Medium   | < 2 hari kerja   | < 10 hari kerja   |
| Low      | < 5 hari kerja   | < 30 hari kerja   |

### 15. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

### 16. APPENDICES

#### Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

## Appendix B: Domain-Centric Governance

| Domain              | Data Owner               | Steward         | Custodian        |
|---------------------|--------------------------|-----------------|------------------|
| Customer            | Head of CRM/Marketing    | CRM Steward     | Data Platform/IT |
| Product             | Merchandising Lead       | Catalog Steward | IT               |
| Order & Transaction | Operations/Payments Lead | Order Steward   | IT               |
| Vendor              | Supply Chain Lead        | Vendor Steward  | IT               |

## Appendix C: Governance Cadence

| Cadence   | Forum                    | Purpose                                              |
|-----------|--------------------------|------------------------------------------------------|
| Monthly   | Domain Governance Review | DQ trends, open issues, access exceptions            |
| Quarterly | Governance Council       | Policy decisions, cross-domain conflicts, KPI review |
| Annually  | Policy/TOM Review        | Regulatory changes, roadmap reprioritization         |

## Appendix B: Sample Workflow

- Data Issue Management Workflow Diagram

# Visual Diagram (Architecture + Governance Flow)

